

Tale of Two Circuits: A Meta-Analysis Comparing the Efficacy of Deep Brain Stimulation for Parkinson's Disease versus Treatment-Resistant Depression

Seyanne Govender, University of Essex, sg26761@essex.ac.uk

ABSTRACT

Deep Brain Stimulation (DBS) is an advancing neurosurgical procedure that involves implanting electrodes to deliver targeted electrical impulses to specific regions of the brain, offering a powerful therapeutic option for otherwise treatment-refractory conditions (Sremic et al., 2025). It targets two neural systems: the basal ganglia-thalamocortical motor circuit in Parkinson's Disease (PD), and the cortico-limbic affective circuit in Treatment-Resistant Depression (TRD), involving network-wide dysregulation (Holtzheimer & Mayberg, 2011). While DBS is an effective therapy for motor symptoms in PD (Perestelo-Pérez et al., 2014), its use in TRD has yielded varied outcomes, highlighting a need to understand its effects on specific circuits (Hovington et al., 2021). Despite extensive research within each disorder, a direct comparative synthesis on how pathophysiology influences therapeutic efficacy is lacking. This proposal outlines a meta-analysis and narrative review to address this gap. The primary aim is to compare the clinical efficacy and neurophysiological effects of DBS on PD motor circuits versus TRD affective circuits. By integrating clinical evidence with neurobiological mechanisms, this research will provide an evidence-based perspective on DBS's therapeutic scope. The findings are expected to inform future clinical applications, refine patient selection criteria for TRD, and guide research by elucidating the relationship between target circuit pathology and therapeutic outcome.

INTRODUCTION

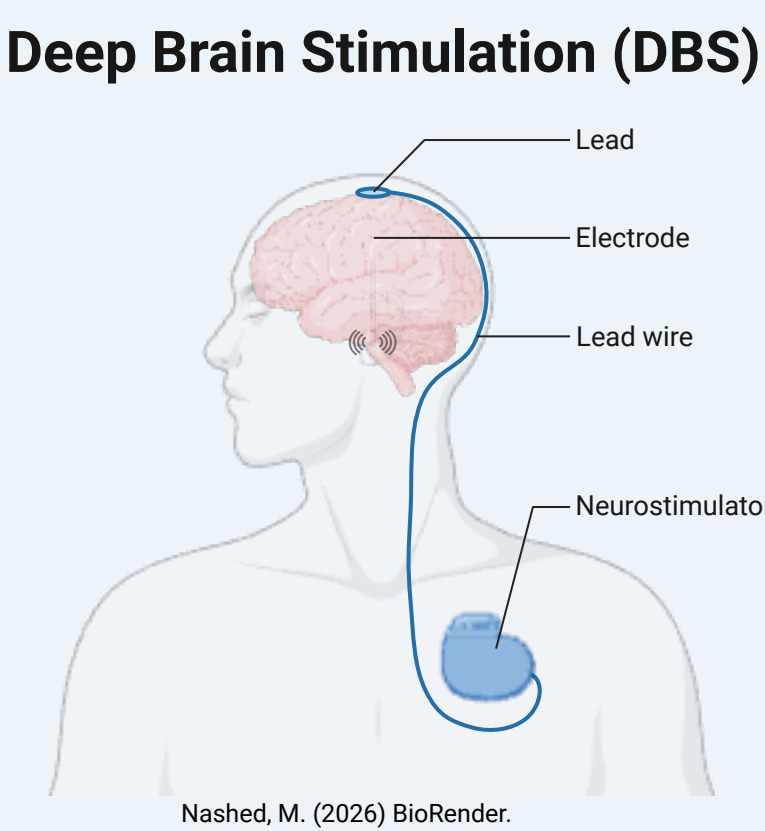
DBS was developed for movement disorders at first (Deuschl et al.,2006; Sremic et al., 2025). The success of DBS in PD is founded on a relatively well-understood pathophysiology: the progressive and focal loss of dopaminergic neurons in the substantia nigra leads to pathological hyperactivity within the basal ganglia-thalamocortical motor circuit (Kalia & Lang, 2015). In PD, the body of evidence is robust; meta-analyses of randomized controlled trials consistently demonstrate a significant and reliable effect size for reducing motor symptoms(Perestelo-Pérez et al., 2014). In stark contrast, the application of DBS in TRD, has yielded a more complex and challenging clinical scenario. Treatment-Resistant Depression (TRD), a debilitating condition whereby patients do not respond to various conventional therapies such as medication and psychotherapy (Voineskos et al., 2020), is believed to involve more diffuse, network-wide dysregulation across a cortico-limbic affective circuit rather than a single, localised lesion (Holtzheimer & Mayberg, 2011).

While seminal studies targeting nodes such as/like the subcallosal cingulate gyrus (SCC) and ventral capsule/ventral striatum (VC/VS) demonstrated that DBS could produce notable antidepressant effects in some patients (Mayberg et al., 2005; Dougherty et al., 2015), subsequent research has been marked by significant outcome variability. Although some patients experience significant and lasting remission, other patients show only a fragmentary response or no benefit. Recent systematic reviews and meta-analyses, corroborate this heterogeneity, indicating more modest mean effect sizes and wider confidence intervals compared to those observed in PD (Hovington et al., 2021; Scangos et al., 2021). This discrepancy in clinical outcomes between PD and TRD highlights a critical and unaddressed gap in the neuromodulation literature.

The primary aim is to conduct a meta-analysis comparing the clinical efficacy and neurophysiological effects of Deep Brain Stimulation on the basal ganglia-thalamocortical circuits in Parkinson's Disease versus the cortico-limbic circuits in Treatment-Resistant Depression.

It is hypothesized that the quantitative meta-analysis will reveal a significantly larger and more consistent pooled effect size for DBS in treating the motor symptoms of PD compared to the depressive symptoms of TRD. The additional hypothesis is that a narrative synthesis of the existing neuroimaging and electrophysiological data will reveal that the difference in clinical efficacy uncovered by the met-analysis, will be explained by the underlying contrast in circuit pathology, with the focal nature of dysfunction in PD being more amenable to a targeted intervention than the distributed network dysfunction characteristic of TRD.

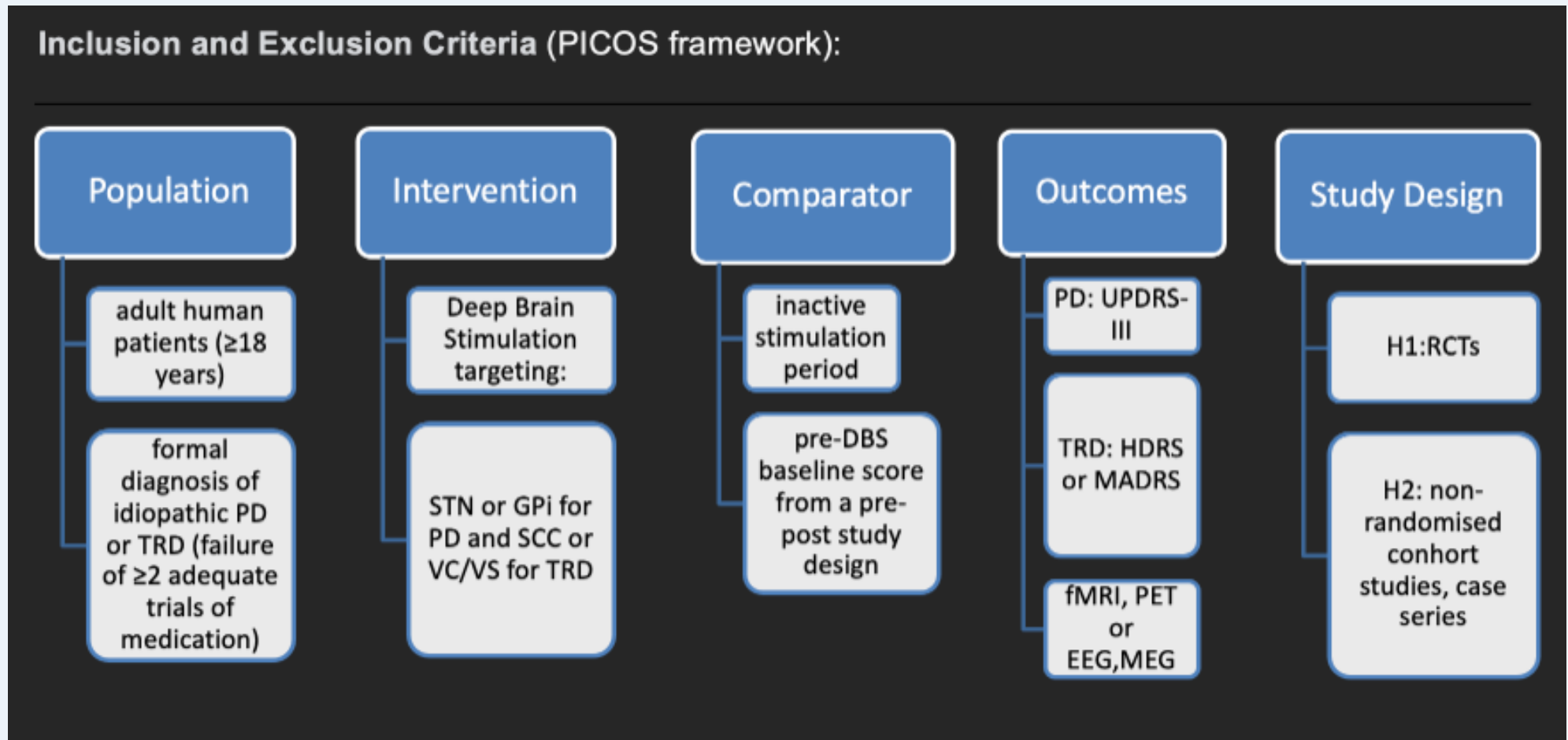
METHODS



This research exclusively involves the analysis of previously published and publicly available data. Consequently, direct ethical approval from an institutional review board is not required, as no human or animal participants will be directly involved. It is important to note that this study will not involve accessing or analysing any confidential or individual patient-level data. The integrity of patient anonymity from the primary studies will be fully maintained, as all data will be extracted in its aggregated form as reported in the source publications. The primary ethical obligation is to ensure the transparent, accurate, and unbiased synthesis and reporting of the findings from the original studies. This duty includes accurately representing the work of the primary researchers as well as honouring the contributions of the patients involved in those studies. The full study protocol will be pre-registered on the International Prospective Register of Systematic Reviews (PROSPERO) to uphold this commitment to methodological transparency and mitigate the risk of reporting bias.

Two independent reviewers will extract the relevant data using a standardized data extraction form. The primary quantitative analysis will be conducted using statistical software such as the 'metafor' package in R or Review Manager (RevMan 5.4). Standardized mean differences (SMDs) will be calculated for clinical outcomes to allow for comparison across different rating scales. These will be pooled using a random-effects model to account for the anticipated clinical and methodological heterogeneity between studies. For the neurophysiological data, a formal quantitative meta-analysis is not feasible due to methodological diversity; therefore, a narrative synthesis will be conducted. This will involve systematically summarizing and thematically analyzing the reported changes in network activity to identify convergent patterns of circuit modulation within and across the two disorders.

This study will be a meta-analysis (Phase 1) followed by a narrative review (Phase 2) derived from the meta-analysis findings. This study will be rigorously structured and reported in accordance with the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) 2020 statement (Page et al., 2021). To ensure methodological transparency, prevent duplication of research, and pre-specify our analytical plan to mitigate reporting bias, the full protocol for this review will be pre-registered with the International Prospective Register of Systematic Reviews (PROSPERO). Phase 1 will be a quantitative meta-analysis conducted on data pertaining to clinical outcomes to both compare and calculate the pooled effect sizes of DBS on motor symptoms in PD using the Unified Parkinson's Disease Rating Scale Part III (UPDRS-III) and depressive symptoms in TRD using Hamilton Depression Rating Scale (HDRS) or the Montgomery-Åsberg Depression Rating Scale (HDRS/MADRS) and facilitate a direct statistical comparison of the therapeutic impact. Phase 2 will be a narrative synthesis of electrophysiological and neuroimaging data from included studies, conducted to both identify and compare the network-level effects of DBS in each disorder.



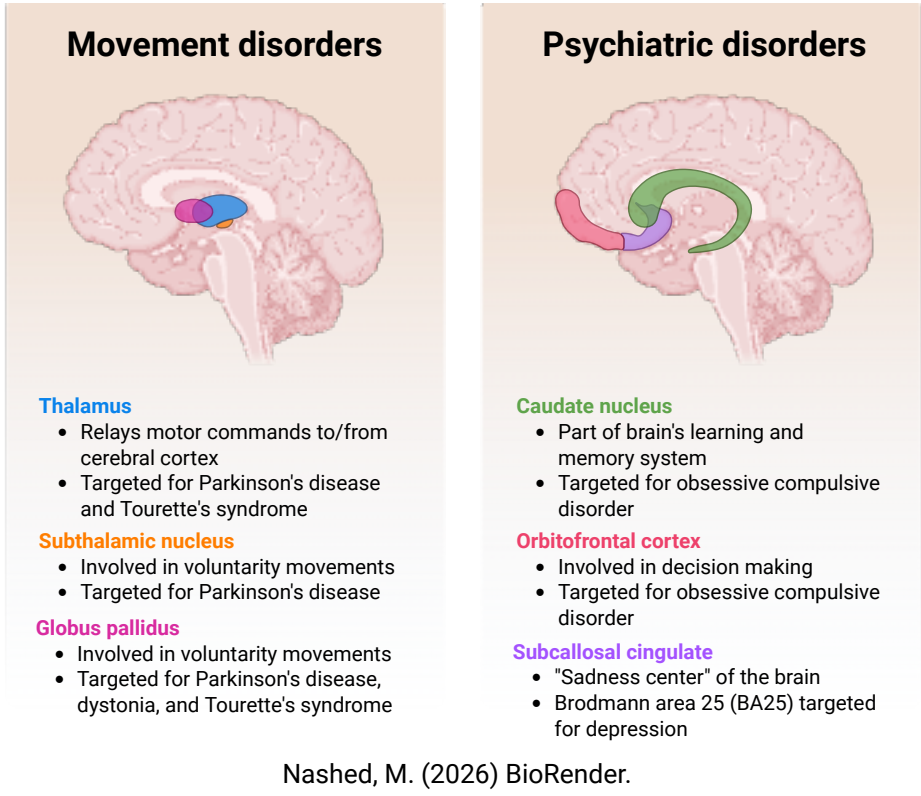
A comprehensive and systematic literature search will be conducted across the PubMed (MEDLINE), Embase, PsycINFO, and Web of Science databases. Following the search, all retrieved citations will be managed using systematic review software (e.g., ASReview) for screening. To minimize selection bias, two independent reviewers will screen all titles and abstracts, followed by a full-text review of potentially eligible articles against the pre-specified inclusion criteria. Any disagreements throughout this process will be resolved by consensus or, if necessary, by a third independent reviewer.

LIMITATIONS

A primary challenge for this meta-analysis will be addressing clinical heterogeneity across included studies (Higgins et al., 2022). Variability in patient populations (e.g., disease duration, symptom severity) and DBS programming parameters is expected, potentially affecting pooled effect sizes. This is crucial for TRD, a clinically heterogeneous condition with diverse biotypes (Voineskos et al., 2020). Such variance can interfere statistically and reduce precision. To address this, we will conduct subgroup analyses and meta-regression, given sufficient data, to examine variable influence on treatment outcomes. Methodological heterogeneity will also arise from neuroimaging and electrophysiological techniques in source studies. As these methods cannot be quantitatively compared, we chose a narrative synthesis to integrate findings without overstating statistical congruence. Additionally, our findings' validity may be limited by publication bias, where studies with positive findings are more published (Sterne et al., 2011). This bias could inflate DBS treatment effects. We will assess this risk by evaluating funnel plot symmetry and using statistical methods, like Egger's regression test, to detect small study effects suggesting bias (Egger et al., 1997). Finally, our conclusions' generalisability will be limited by the scope of existing literature. Findings will stem from specific patient populations and intervention methods in published RCTs, which often apply strict criteria, limiting external validity (Rothwell, 2005). Thus, findings may not fully represent all DBS patients, especially those with comorbidities or from underrepresented groups. While this limitation is unavoidable, this synthesis will provide a critical evidence-based efficacy benchmark under controlled conditions and highlight literature gaps for future research.

CONCLUSION

The "connectomic approach," uses brain imaging to identify and map these circuits. This approach is being used more frequently to guide DBS surgery, highlighting a shift towards a network-based understanding in this field (Riva-Posse et al., 2018). Given this, the proposed research is not merely a clinical comparison; it is a foundational inquiry into how a neuromodulatory intervention interacts with the different pathophysiological characteristics of distinct neural circuits. This proposed meta-analysis and narrative review will directly address this gap by quantitatively comparing treatment efficacy and synthesizing neurophysiological findings from DBS studies in both PD and TRD. Despite this large body of research conducted separately within each disorder, a significant gap in the literature remains: no study has systematically synthesised the available evidence to directly and quantitatively compare the efficacy of DBS across these two distinct disorders. Applying DBS to both PD and TRD translates to a real-world test of this theory, seeing that it uses a single technology to modify two foundationally distinct neural circuits - one primarily motor, the other affective, subsequently raising a discerning question regarding the comparative efficacy of a single technology across different pathological domains.



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